[redacted]

6/16/2025

ok we are going into L^1 today. a non-Euclidean metric. we recall that this is a metric given by $d(A,B) = \sum_i |b_i - a_i|$, where the a_i, b_i are coords of the two points. this sure is a valid metric cuz it satisfies the three things: symmetry, positive definiteness, triangle inequality. what I want to do today, however, is arguably not as abstract as the other contexts in which we've seen this norm: we are gonna do geometry under the L^1 metric.

let us start in the plane. if we're playing with a new distance, then we ought to consider certain loci of points. for example, the collection of all points equidistant from a given point w/ a given equidistance—

that's a circle, and as we know a circle is a square in L^1. aside from visual divergence, you can note that two circles can intersect in more than 2 places; ie, an entire side, or maybe two! so now we consider the "perp" bisector of a line segment between two points. because of the weird circle interactions, there are THREE ways we can get a perp bisector. this is weird and cool. but other loci get better.

locus of pts equidistant from 2 given pts

here are the cases of the ellipse, parabola, & hyperbola; loci of points $\{P \in \mathbb{R}^2 \mid d(P,A)+d(P,B)=c\}$, $\{P \in \mathbb{R}^2 \mid d(P,A)=d(P,\ell)\}$, $\{P \in \mathbb{R}^2 \mid |d(P,A)-d(P,B)|=c\}$, respectively. they all have many cases!

Sure, we can also ascribe these to circle behavior, but I think we all know these last 3 as conic sections. ok, now we graduate to thinking in 3D: what is a cone under L^1? a square pyramid! (I mean, pyramids are just like discretized cones.) now the behavior is even clearer. pyramids are surely less symmetric than cones; so we get all these different-looking parabolas.

Dandelin spheres sorta

wait a second, this doesn't cover the hex or octa ellipses, or the infinite-plane-section hyperbolas. ok, so that's coz in L^1, the distance rules above for defining these guys is NOT equivalent to the focus, directrix, eccentricity definition. L^1 is violating a lot of Euclidean intuition!

foci thingy is not related to OG "cone" of conic sections anyway

just for safety, let's look carefully at the Euclidean axioms. (actually, these are Birkhoff's equivalent formulations, which are more comprehensive to work with to actually

$e = \dfrac{\cos\alpha \leftarrow \text{plane to axis}}{\cos\beta \leftarrow \text{cone surface to axis}}$

construct & look at stuff.) incidence: 2 pts give a line, and ∃ 3 non collinear pts. measure: there is a bijection between a line and ℝ. separation: you can make rays from a line. angle measure: two rays w/ common end point but not common line have an angle between them, arbitrarily given to range from 0 to π. alternatively, the set of rays from a given point have a bijection with ℝ (mod 2π) (also arbitrary modulo). SAS similarity. parallel axiom. take a wild guess which one does not hold in L' (NOT the angle one, that is a can of worms we will open next). while you're thinking, I will point out that your first (likely first, I mean) non-Euclidean space was hyperbolic geometry, which was off by the parallel axiom: given a line and a point not on the line, there are AT LEAST TWO lines going through said point that do not intersect said line.

WE WILL RECONSIDER IN A BIT

so, enough suspense: SAS fails. so do most triangle congruences. only if you have 3 sides and 2 angles - the strongest requirements! - can you be guaranteed congruence in L'.

P2 to show counter example (rotation)

in my example, you can notice that basically the triangle got "rotated", so now its angles aren't the same. honestly, this is the crux of all the funny deals with L', from the varying bisectors to the hint hint wink wink earlier: due to much more limited symmetry, L' is NOT a rotation-invariant space! in fact, we talked about the isometries of the plane under L' - only rotations by 90° preserve things!

well, isn't that fishy? maybe, some of you are saying, if it's not rotation-invariant, L^1 geometry shouldn't see angles the same way?
at least 1 paper I found certainly thinks so. like, if we try to define π as "ratio of circumference to diameter," we get 4 in L^1! ok, and since we can't do inner products, let's stick with this unit circle definition. one t-radian is an angle that intercepts a unit circle arc of length 1. since from the + x-axis this looks like 45° in Euclidean measure, we can convert between the two types of angles by

θ = distance from (1, 0) to intersect $y = -x + 1$, $y = x \tan_e \theta_e$

$$\theta = 2 - \frac{2}{1 + \tan_e \theta_e}$$

where the e subscript means Euclidean. in terms of

6/17/2025 we can define sin & cos the same way: as horizontal & vertical projections of an angle's endpoint onto x, y axes. as you may expect from the shape of the circle, the sine & cosine are non-jagged, too! piecewise.
you can re-derive all the trig formulas, as long as you go back to the unit circle def'n for angles; anything you might rely on SAS (or — or indeed, anything relying on a triangle — won't work (e.g. the sum-of-angles formulas).
similar to angle conversion, there is a distance conversion formula (which I could have given you at the start of the talk, but then there would be no talk):

$$d(A, B) = \frac{\sqrt{1 + m^2}}{1 + |m|} d_T(A, B), \quad m = \frac{\Delta y}{\Delta x}$$

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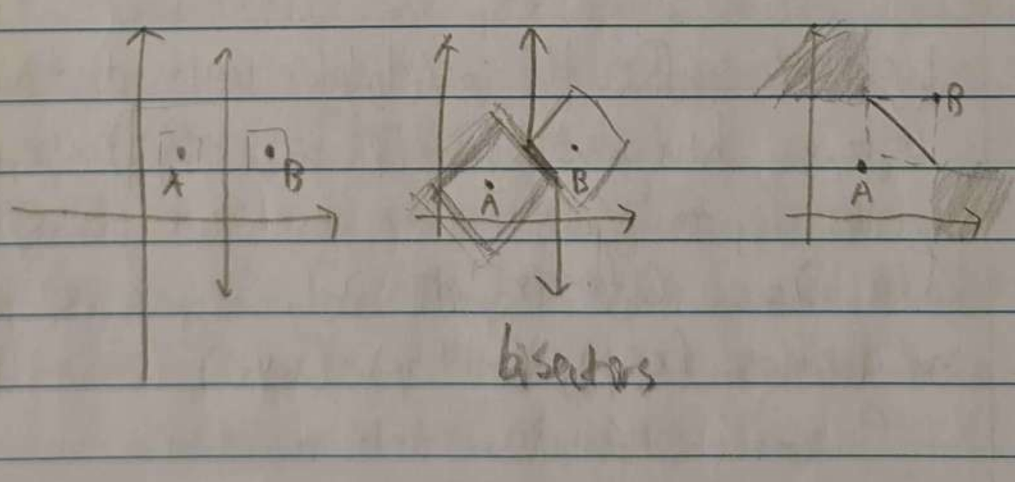
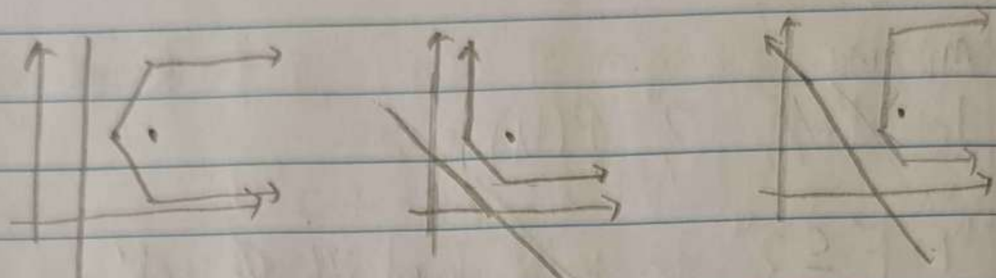
6/17/2025 MORE LEC NOTES

and you can use this to also derive area formulas for any polygon, as well as the pythagorean theorem, which btw, ftr:

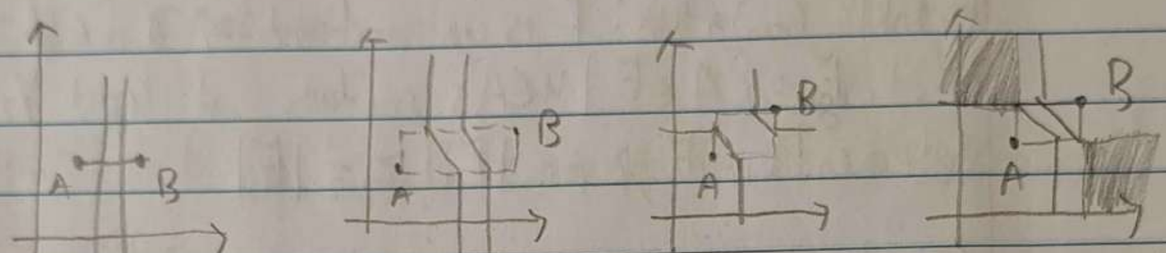
$a = b + c$ (legs parallel to axes)
$(1+m)a = (bm+c) + |cm-b|$ (any other triangle)





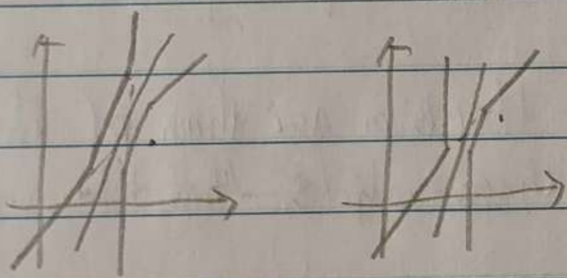
← do this by hand

parabolas



$s=0$ $\quad$ $s<(D-k)/2$ $\quad$ $s>(D-k)/2$ $\quad$ $s=(D-k)/2$

hyperbolas
(for constant c, changing short side of box)

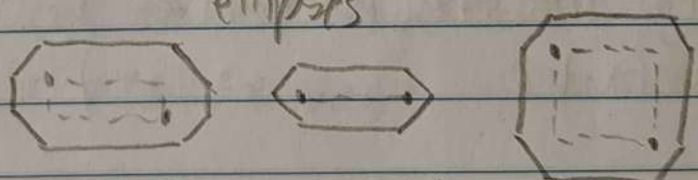


focus-directrix:

$|m|<e$ $\quad$ $|m|=e$

↑
directrix slope

ellipses



focus-directrix: all quads

SAS failure: